

# Sumit Kumar Panda

(848)667-2576|North Brunswick, NJ-08902|sumitk.panda94@gmail.com|[LinkedIn](#)

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## SUMMARY

- Experienced in Data Engineer with 6+ years of experience in designing, developing and implementing scalable solutions.
- Proficient in various AWS services along with provisioning and maintaining using AWS resources such as EMR, S3 buckets, EC2 instances, RDS, CloudWatch, RedShift and data ingestion with AWS Step Functions, AWS Glue.
- Proficient in various programming languages and big data technologies, including Python, Hadoop, Spark, Map Reduce and Hive.
- Skilled in cloud platforms such as AWS, with experience and implementing secure and scalable data solutions on these platforms.
- Hands on experience in real time data streaming solutions using Apache Spark, Spark SQL & Data Frames and Spark streaming.
- Proficient in importing and exporting data between HDFS and Relational system using tools like Sqoop.
- Proficient in data migration, transformation, integration and knowledge of data modeling concepts such as Star-Schema Modeling, Snowflake Schema Modeling, and fact and Dimension tables.
- Strong understanding of Agile Methodologies and experience working in an Agile development environment.

## EXPERIENCE

### Data Engineer-I Amazon

Jan. 2023 – Nov. 2024  
New York, NY

- Successfully migrated on-prem ETL workloads to AWS using a combination of DMS, S3, Glue, Redshift, and Athena. This resulted in a 40% increase in efficiency and scalability, allowing the organization to handle larger volumes of data while maintaining optimal performance.
- Leveraged AWS Lake Formation to enable fine-grained access-controlled reporting and analytics for business users. This empowered stakeholders to derive meaningful insights from data while ensuring data security and compliance.
- Developed and implemented detailed data security protocols, utilizing AWS KMS and IAM, to safeguard sensitive company data. These protocols enhanced data protection and ensured that only authorized personnel could access critical information.

- Implemented Athena data sources on S3 buckets to facilitate adhoc querying and business dashboarding. Leveraging Amazon Quicksight, I provided business partners with intuitive data visualization tools, enabling them to easily make data-driven decisions.
- Designed and implemented data models and data warehousing solutions using AWS services such as Redshift and Athena.
- Optimized EMR clusters by improving cluster stability and performance by fine tuning resource allocation within the EMR cluster, ensuring optimal utilization of resources based on Workload patterns.
- Fine-tuned Spark configurations by deploying improved parallelism configurations in EMR clusters, reducing load times for some of our larger datasets by 25-30 minutes.
- Collaborating closely with stakeholders to gather requirements, business analysis, Design Review and Development, Testing.
- Creating comprehensive consulting proposals that address specific customer requirements, ensuring alignment with AWS best practices and solutions.

**Data Engineer**  
**Shutterfly, INC**

**Feb. 2022 - Jan. 2023**  
**New Jersey, NJ**

- Designed and developed scalable ETL pipelines using AWS Glue, Apache Airflow, and custom Python scripts to ingest, transform, and load data into AWS S3, optimizing data flow and ensuring real-time and batch processing efficiency.
- Integrated on-premises data systems with cloud data architectures, enabling seamless data migration and hybrid data solutions. Used custom API calls to extract and transfer on-prem data to AWS S3, ensuring secure and reliable data flow across systems.
- Ingested transactional data from two primary sources using Spark, S3, Redshift, and Python. This streamlined the data ingestion process and improved the timeliness of data availability for critical business operations.
- Identified transactional systems (SOR - System of Records) and implemented business data lineage and transformation for KPI reporting. This allowed stakeholders to track data origins and transformations, ensuring data integrity and supporting informed decision-making.
- Conducted data analysis, employing various techniques to extract, cleanse, and manipulate data from diverse sources providing accurate and reliable data that supported business decision-making processes and helped drive operational efficiency.

- Developed and maintained reporting dashboards in Looker, enabling stakeholders to track key performance indicators (KPIs) and other business metrics. Furthermore, I optimized Looker connections to Redshift, significantly reducing report refresh times to 1-2 minutes.
- Automated resulting scripts and workflow using Apache Airflow and Shell scripting to ensure daily execution in production.
- Implemented monitoring and alerting systems using Cloudwatch to ensure high availability and performance of data services.
- Developed dashboards and visualizations to help business users analyze data as well as providing data insight to upper management with a focus on Microsoft products like SQL Server Reporting Services and Power BI.

**Data Engineer**  
**L&T Infotech**

**March 2017 - Dec. 2020**  
**Pune, MH, India**

- Utilized Spark SQL/Hive to implement business transformations and perform anomaly detection for credit card transactions. This played a critical role in fraud detection and prevention, safeguarding the financial interests of clients.
- Ingested data from multiple sources using Sqoop into HDFS and performed transformations using Hive and Spark. This allowed for efficient processing and analysis of large volumes of data, resulting in improved data insights and decision-making.
- Implemented partitions and bucketing techniques in Hive, optimizing table performance and reducing query response times. By fine-tuning the data storage layer, I significantly enhanced overall system efficiency.
- Designed and developed ETL workflows using Oozie, automating data processing tasks and reducing manual intervention. This increased productivity and ensured consistent and reliable data processing.
- Collaborated closely with clients to understand their unique business needs and translate them into practical and effective data processing applications. By aligning technology solutions with business requirements, I delivered tailored solutions that added significant value to clients' operations.
- Implemented data lake (Cloudera) to store data sourced from SQL Server into HDFS and built data pipeline to process raw data set into transformed data for downstream use.
- Built a robust data pipeline to transform raw datasets into publishable formats using the Conformance engine. By implementing transformations such as transforming XML data into Parquet, I ensured the data was optimized for downstream consumption.

- Leveraged my expertise in Unix and developed modules to automate the entire batch process. This automation not only saved valuable time but also improved the overall reliability and accuracy of data processing.
- Maintained and configured the extracted data using Zookeeper and tracked the nodes of the high-volume data.
- Implemented MapReduce programs to process semi-structured data formats like XML and JSON.
- Performed business data analysis and report creation in support of both ad hoc requests from the customer and to support the developments and visualized the data and formed reports using the visualization tools like Tableau.

## SKILLS

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- ❑ **Cloud:** AWS DMS, S3, Glue, EMR, Cloudwatch, Airflow, Redshift, Athena, Lake Formation, Quicksight, RDS, IAM, KMS
- ❑ **Big Data:** Hadoop, HDFS, Hive, Spark, Sqoop, Oozie, Map Reduce, Zookeeper
- ❑ **Database:** Oracle, MySQL, Teradata
- ❑ **Programming Languages:** Python, PySpark, Unix Shell Scripting
- ❑ **Data Visualization & Tools:** Tableau, PowerBI
- ❑ **Operating Systems:** Linux, Windows, Mac OS, Unix

## EDUCATION

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**Cleveland State University, Cleveland, OH**  
Master of Information System

**Graduated: May 2022**  
GPA: 3.83/4.0

**Trident Academy of Technology, Odisha, India**  
Bachelor's in Electrical Engineering

**Graduated: Mar. 2016**  
GPA: 7.5/10.0

## CERTIFICATION

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**AWS Certified Solutions Architect - Associate**